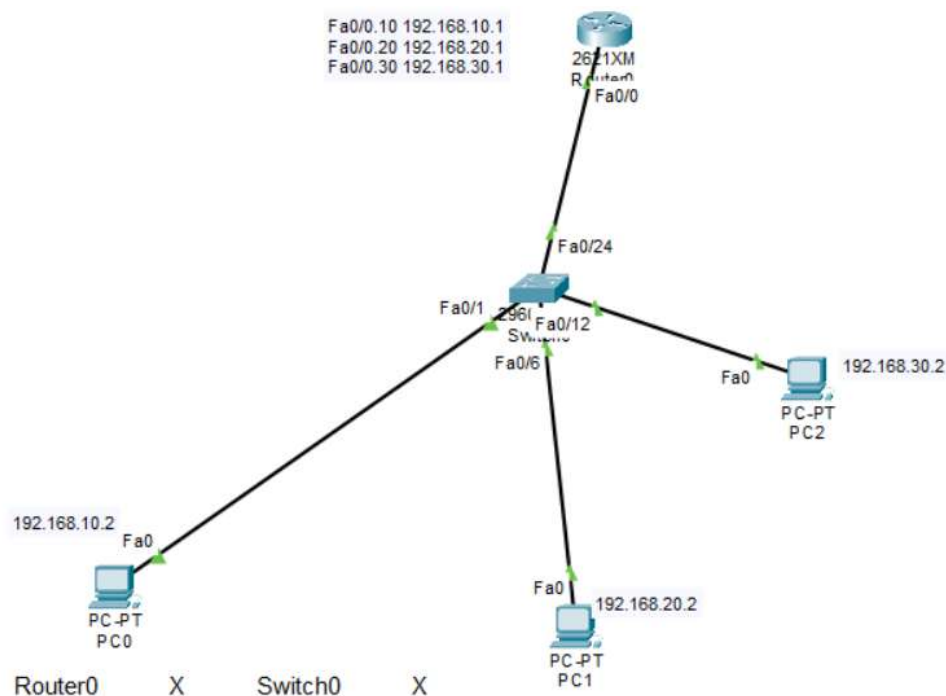


Inter-VLAN Routing Concepts

Task 1: On a Cisco Packet Tracer or GNS3 simulation, create three VLANs named Insta, Zomato, and Flipkart on a single switch, and assign at least one PC to each VLAN.

Three VLANs were successfully configured on Switch0 and specific access ports were assigned to each VLAN broadcast domain as per the requirements:

- VLAN 10 (Insta) — Ports FastEthernet 0/1 to 0/5 | PC0 IP: 192.168.10.2/29
- VLAN 20 (Zomato) — Ports FastEthernet 0/6 to 0/10 | PC1 IP: 192.168.20.2/29
- VLAN 30 (Flipkart) — Ports FastEthernet 0/11 to 0/15 | PC2 IP: 192.168.30.2/29



Switch Configuration Commands Used:

```
Switch# configure terminal
Switch(config)# vlan 10
Switch(config-vlan)# name Insta
Switch(config-vlan)# exit
Switch(config)# vlan 20
Switch(config-vlan)# name Zomato
Switch(config-vlan)# exit
Switch(config)# vlan 30
Switch(config-vlan)# name Flipkart
```

```
Switch(config-vlan)# exit
```

```
Switch(config)# interface range fastEthernet 0/1 - 5  
Switch(config-if-range)# switchport mode access  
Switch(config-if-range)# switchport access vlan 10  
Switch(config-if-range)# exit
```

```
Switch(config)# interface range fastEthernet 0/6 - 10  
Switch(config-if-range)# switchport mode access  
Switch(config-if-range)# switchport access vlan 20  
Switch(config-if-range)# exit
```

```
Switch(config)# interface range fastEthernet 0/11 - 15  
Switch(config-if-range)# switchport mode access  
Switch(config-if-range)# switchport access vlan 30  
Switch(config-if-range)# exit
```

Task 2: Configure a Router-on-a-Stick setup by creating sub-interfaces (e.g., FastEthernet0/0.10, .20, .30) on a simulated router, each with an IP address for the corresponding VLANs Insta, Zomato, and Flipkart.

Sub-interfaces were created on Router0 under the main physical interface FastEthernet0/0, configured with 802.1Q encapsulation for matching VLAN tags, and assigned IP addresses to serve as Default Gateways:

Router Configuration Commands Used:

```
Router# configure terminal
```

```
Router(config)# interface fastEthernet 0/0
```

```
Router(config-if)# no shutdown
```

```
Router(config-if)# exit
```

```
Router(config)# interface fastEthernet 0/0.10
```

```
Router(config-subif)# encapsulation dot1Q 10
```

```
Router(config-subif)# ip address 192.168.10.1 255.255.255.0
```

```
Router(config-subif)# exit
```

```
Router(config)# interface fastEthernet 0/0.20
```

```
Router(config-subif)# encapsulation dot1Q 20
```

```
Router(config-subif)# ip address 192.168.20.1 255.255.255.0
```

```
Router(config-subif)# exit
```

```
Router(config)# interface fastEthernet 0/0.30
```

```
Router(config-subif)# encapsulation dot1Q 30
```

```
Router(config-subif)# ip address 192.168.30.1 255.255.255.0
```

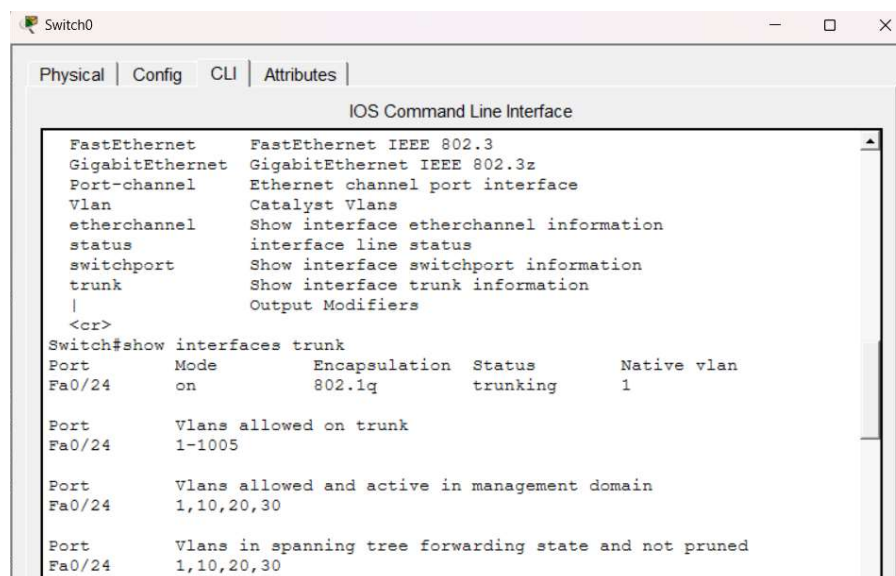
```
Router(config-subif)# exit
```

Task 3: Assign the switch port connected to the router as a trunk port, and verify trunking is active using the show interfaces trunk command.

The switch port FastEthernet 0/24 connected to the router's physical interface FastEthernet 0/0 was configured as an 802.1Q trunk link.

Switch Configuration Commands Used:

```
Switch# configure terminal
Switch(config)# interface fastEthernet 0/24
Switch(config-if)# switchport mode trunk
Switch(config-if)# exit
```



Task 4: Test inter-VLAN connectivity by pinging from a PC in the Insta VLAN to a PC in the Zomato VLAN, and explain why the ping succeeds or fails based on your configuration.

- Ping Test Result:

Pinging from PC0 (VLAN 10 - 192.168.10.2) to PC1 (VLAN 20 - 192.168.20.2) resulted in 0% packet loss (4/4 replies received successfully).

- Detailed Explanation:

VLANs operate at Layer 2 and inherently isolate broadcast domains, preventing direct communication between devices on different VLANs. In this Router-on-a-Stick setup, when PC0 sends a packet to PC1 (192.168.20.2), PC0 recognizes that the target IP belongs to another subnet and forwards the packet to its default gateway (192.168.10.1).

The packet leaves Switch0 on trunk port Fa0/24 tagged with VLAN ID 10. Router0 receives the frame on sub-interface Fa0/0.10, strips the 802.1Q tag, consults its internal IP routing table, and routes the packet to sub-interface Fa0/0.20. Router0 then encapsulates the IP packet into an Ethernet frame tagged with VLAN ID 20 and sends it back down the same trunk link to Switch0, which forwards it to PC1. This enables seamless Inter-VLAN routing.

Task 5: Refactor your Router-on-a-Stick configuration to add a new VLAN called 'Spotify' without disrupting existing VLAN communication, and document the exact commands you used for both the switch and the router.

The configuration was extended to include VLAN 40 (Spotify) non-disruptively while preserving existing routing for VLANs 10, 20, and 30.

Exact Switch Commands Used:

```
Switch# configure terminal
Switch(config)# vlan 40
Switch(config-vlan)# name Spotify
Switch(config-vlan)# exit

Switch(config)# interface range fastEthernet 0/16 - 20
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 40
Switch(config-if-range)# exit
```

Exact Router Commands Used:

```
Router# configure terminal
Router(config)# interface fastEthernet 0/0.40
Router(config-subif)# encapsulation dot1Q 40
Router(config-subif)# ip address 192.168.40.1 255.255.255.0
Router(config-subif)# exit
```

